

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Winter Examination – 2022

Course: B. Tech. Branch : Information Technology Semester : V

Subject Code & Name: BTITPE503A & Embedded System

Max Marks: 60

Date: 02/02/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1	Solve Any Two of the following.		12
A)	What are common design metrics? List a pair of design metrics that may compete with one another.	L1/CO1	6
B)	List the different processor technologies. Compare what are the benefits of using each processor technologies?	L3/CO1	6
C)	Draw and explain ARM core dataflow model.	L2/CO1	6
Q.2	Solve Any Two of the following.		12
A)	Summarize flow of actions of interrupt- driven I/O using fixed ISR address.	L2/CO1	6
B)	What is I2C? Draw and explain bus structure of I2C protocol	L2/CO1	6
C)	Write a short notes on i) PCI bus ii) IrDa iii)USB	L1/CO1	6
Q. 3	Solve Any Two of the following.		12
A)	What is FSM? Describe basic state machine model for elevator control system	L3/CO1	6
B)	Explain concurrent process model using simple example?	L2/CO1	6
C)	What are the advantages and disadvantages of embedded programming in C++?	L2/CO1	6
Q.4	Solve Any Two of the following.		12
A)	What are services provided by Kernel in operating system?	L1/CO2	6
B)	Explain about RTOS μ COS-II in brief.	L1/CO2	6
C)	Give basic features of RTOS VxWorks.	L1/CO2	6

Q. 5	Solve Any Two of the following.		12
A)	Draw and explain hardware architecture of digital camera.	L2/CO2	6
B)	What are the requirements of TCP/IP network?	L2/CO2	6
C)	How Smart card explained using class diagram?	L3/CO3	6

***** End *****